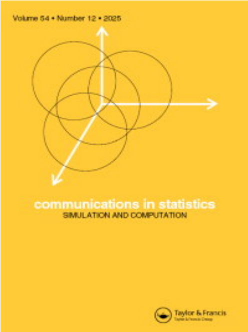

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Communications in Statistics - Simulation and Computation

ISSN: 0361-0918 (Print) 1532-4141 (Online) Journal homepage: www.tandfonline.com/journals/lssp20

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COMMUNICATIONS IN STATISTICS - SIMULATION AND COMPUTATION®
2025, VOL. 54, NO. 12, 5065–5078
https://doi.org/10.1080/03610918.2024.2430732

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ABSTRACT

In this study, we focus on visualizing trajectory data, a type of data based on a series of temporal observations. We present an adapted version of t-distributed stochastic neighborhood embedding (t-SNE) tailored for trajectory data. This method is designed to preserve the inherent curved structure of trajectory data by incorporating a robust distance measure. Furthermore, it demonstrates the ability to maintain the data structure even in the presence of missing values at different time points. The performance of the proposed method is rigorously evaluated through a simulation study and demonstrated its effectiveness in visualizing two types of trajectory data, Gait data and NBA data.

ARTICLE HISTORY
Received 4 March 2024
Accepted 11 November 2024

KEYWORDS
Dynamic time warping; t-Distributed stochastic neighborhood embedding; Trajectory data; Visualization

1. Introduction

Trajectory data is a high-dimensional tensor data that observes the movement or path of an object over time. With advancements in modern computer technology, the accessibility to such has significantly increased across various fields. However, these data often pose challenges, including missing values and the complexity of (multidimensional) trajectories, particularly when recorded in large-scale data formats.

For instance, data measuring the angles of movement for the hips, knees, and ankles of 28 individuals walking on a treadmill (Tanghe et al. 2020) is a multivariate trajectory data consisting of six variables, accounting for both left and right sides. Another example involves tracking of GPS routes of 1000 taxis in and around the Manhattan area over a year (Benson, Gleich, and Lim 2017). This data encompasses long-term recorded large-scale trajectory data, including pick-up and drop-off locations, taxi IDs, operating times, and more. In the field of biology, trajectory data is observed, such as attaching GPS trackers to oystercatchers to collect data annotated with eight different types of behaviors over a month (Shamoun-Baranes et al. 2012).

Visualization serves as a crucial tool to grasp patterns or trends within trajectory or curve data. As data complexity increases, effective visualization becomes paramount. Over the years, researchers have developed various dimension reduction techniques to visualize high-dimensional data. While Principal Component Analysis (PCA) by Hotelling (1933) is a prominent linear method, it may not be suitable for nonlinear structures like trajectory data. This limitation has led to the development of nonlinear techniques, including Locally Linear Embedding (LLE) (Roweis and Saul 2000), Isomap (Tenenbaum et al. 2000), Stochastic Neighbor Embedding (SNE) (Hinton and Roweis 2002), t-SNE (Maaten and Hinton 2008), and Maximum Variance Unfolding (MVU) (Weinberger, Sha, and Saul 2004). Among them, t-SNE stands out for its ability to capture both local and global relationships.

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However, there is no exploration of t-SNE's use in longitudinal or time series data, and t-SNE relies on Euclidean distance, which is not ideal for measuring the similarity of trajectories. Therefore, it cannot be directly applied to such data. Notably, prior studies lack quantitative assessments of t-SNE's performance by simulation study. To address this gap, our study proposes a visualization approach tailored to the structure and characteristics of high-dimensional trajectory data observed over time. In detail, we utilize a more precise distance measurement – Dynamic Time Warping (DTW). While the conventional t-SNE algorithm uses Euclidean distance to evaluate the similarity between points, this study advocates for using DTW to assess the similarity between trajectories in high-dimensional space. This approach aims to yield more accurate visualization results, better reflecting the inherent patterns within the data. Finally, these efforts establish new possibilities for visualizing and analyzing temporal patterns with t-SNE, with a comprehensive evaluation through simulation studies and real data examples.

The outline of the article is as follows. Section 2 presents trajectory data and provides an overview of DTW distance for trajectory data. Also, we briefly review the original t-SNE. Section 3 introduces the new algorithm embedding DTW into t-SNE. In Sec. 4, we explore the new method through a numerical study, while in Sec. 5, we apply it to two real data examples, Gait data and NBA data. Finally, conclusions are briefly presented in Sec. 6.

2. Existing method

2.1. Data and notation

Let a trajectory data set as $\mathbf{X} = \{\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_n\}$, where $\mathbf{X}_i = \left\{x_i(t_1), x_i(t_2), ..., x_i(t_l)\right\}$ represents the univariate trajectory of the i^{th} subject observed at l time points. On the other hand, for multivariate trajectory data set, $\mathbf{X}_i = \{\mathbf{X}_{i1}, \mathbf{X}_{i2}, ..., \mathbf{X}_{ip}\} \in \mathcal{R}^{p \times l}$ represents the p-dimensional trajectory of the i^{th} subject where $\mathbf{X}_{ij} = \left\{x_{ij}(t_1), x_{ij}(t_2), ..., x_{ij}(t_l)\right\}$ is the j^{th} trajectory observed at l time points. For instance in Gait data, the data observing p joints of n subjects over l time points is a multi-dimensional trajectory data set.

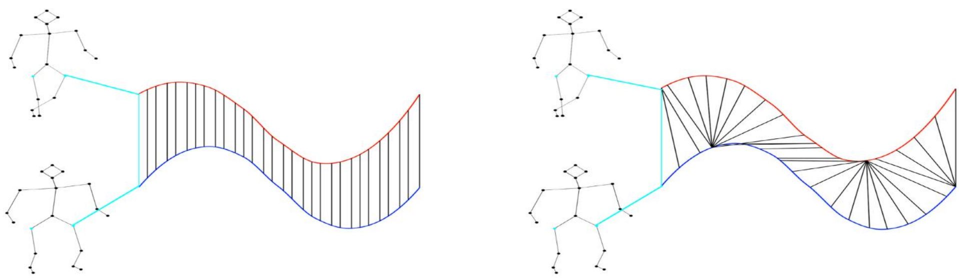
2.2. Dynamic time warping

When considering walking patterns as an trajectory data example, significant variations in walking speed, stride, and movement patterns exist among individuals. Consequently, due to these diverse walking characteristics, the observed trajectories may differ in length for each person.

In such cases, a tool is required to accurately compare and measure the similarity between curves with different lengths and observed points. Traditional methods of measuring similarity, such as Euclidean distance, may yield inaccurate results when comparing patterns due to differences in walking speed. This speed mismatch results in misalignment at the corresponding time points between trajectories, making direct comparisons challenging.

DTW, introduced by Sakoe and Chiba (1978), emerges as a useful tool for comparing trajectories with varying speeds and temporal variations (Ten Holt, Reinders, and Hendriks 2007). DTW seeks an optimal alignment by adjusting the time axes to match points between trajectories, allowing for a more accurate comparison despite differences in speed and timing. This enables a more subtle and accurate evaluation of similarities within various trajectory datasets as illustrated by the gait data example mentioned above.

Figure 1 illustrates the differences between DTW and Euclidean Distance and highlights how DTW's ability to distort the time axis for optimal alignment provides a more accurate comparison between trajectories. In the figure, we compare two trajectories, $\mathbf{X}_i$ and $\mathbf{X}_j$, from a joint in gait data, which have the same length for this example. While Euclidean Distance calculates the distance by directly comparing corresponding points in two sequences, DTW allows for flexible



(a) Euclidean Distance (b) Dynamic Time Warping

Figure 1. Comparison between (a) Euclidean distance and (b) dynamic time warping (DTW) using two trajectories from a joint in gait data.

alignment by considering multiple possible matches between points in one sequence and points in the other sequence. This enables DTW to better handle variations in time or speed between the two trajectories, effectively managing different temporal characteristics.

For the univariate trajectory data, DTW for two curves $\mathbf{X}_i$ and $\mathbf{X}_j$ with different time lengths r and s are defined as follows:

$$DTW(\mathbf{X}_i, \mathbf{X}_j) = [\mathbf{C}]_{r\times s},$$

where C is a $r \times s$ cost matrix and its (k, l)th element is defined as

$$[\mathbf{C}]_{(k,l)} = \sqrt{(X_i(t_k) - X_j(t_l))^2} + min([\mathbf{C}]_{(k-1,l)}, [\mathbf{C}]_{(k-1,l-1)}, [\mathbf{C}]_{(k,l-1)}).$$

As extensions of DTW, two types of multi-dimensional DTW (mDTW), DTW$_I$, and DTW$_D$, are introduced in Shokoohi-Yekta et al. (2017). DTW$_I$ is called the independent DTW which assumes that each variable within multivariate trajectory data is independent. It is simply the sum of univariate DTWs for each variable separately as

$$DTW_I(\mathbf{X}_i, \mathbf{X}_j) = \sum_{q=1}^{p} DTW(\mathbf{X}_{iq}, \mathbf{X}_{jq}).$$

On the other hand, dependent DTW, DTW$_D$, assumes that variables are dependent. Calculated in a similar way to univariate DTW, but redefine the cost matrix C as the cumulative squared Euclidean distance as follows:

$$DTW_D(\mathbf{X}_i, \mathbf{X}_j) = [\mathbf{C}^*]_{r\times s},$$

where $\mathbf{C}^*$ is a $r \times s$ cost matrix and its (k, l)th element is as

$$[\mathbf{C}^*]_{(k,l)} = \sum_{q=1}^{p} (X_{iq}(t_k) - X_{jq}(t_l))^2 + min([\mathbf{C}^*]_{(k-1,l)}, [\mathbf{C}^*]_{(k-1,l-1)}, [\mathbf{C}^*]_{(k,l-1)}).$$

Note that Fréchet distance (FD) introduced by Alt and Godau (1995) is a metric used to measure the similarity between two trajectories or curves. It calculates the similarity by assessing how closely or similarly two trajectories follow each other, considering the actual distance that connects the two trajectories. The FD can be conceptualized as the maximum distance between the two trajectories. Alt and Godau (1995) state that the running time of their algorithm for two polygonal trajectories with m and n segments each is $O(mn \log(mn))$. It is noteworthy that the time complexity of the DTW algorithm is $O(mn)$, indicating that DTW exhibits better computational efficiency compared to FD. Furthermore, DTW allows for temporal distortion, enabling it to measure the similarity between time-biased data effectively. Hence, as the proportion of missing values increases (or the length of curves significantly differs) and the sample size grows, DTW is expected to capture trajectory similarities more effectively than FD.

2.3. t-Distributed Stochastic Neighbor Embedding

The goal of assessing trajectory similarity is for effective visualization of trajectory data. Recently, t-SNE (Maaten and Hinton 2008) has gained attention as a dimension reduction technique that preserves the distances between data points. The technique computes the similarity between data points in the original high-dimensional space as a square Euclidean distance, converting it into a probability distribution for embedding a low dimension.

The joint probabilities p_{ij} in the high-dimensional space by Maaten and Hinton (2008) are defined as

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n},$$

where

$$p_{j|i} = \frac{\exp\left(-\|\mathbf{X}_i - \mathbf{X}_j\|^2 / 2\sigma_i^2\right)}{\sum_{i \neq j} \exp\left(-\|\mathbf{X}_i - \mathbf{X}_j\|^2 / 2\sigma_i^2\right)}. \tag{1}$$

The joint probabilities q_{ij} in low dimension are defined as

$$q_{ij} = \frac{\left(1 + \|\mathbf{Y}_i - \mathbf{Y}_j\|^2\right)^{-1}}{\sum_{k \neq l}\left(1 + \|\mathbf{Y}_k - \mathbf{Y}_l\|^2\right)^{-1}} \tag{2}$$

where σ_i in $p_{i|j}$ is the variance in a Gaussian distribution, and it varies for each trajectory.

Unlike other methods t-SNE considers both local and global structures simultaneously. The parameter responsible for this is perplexity. Perplexity adjusts the number of local neighbors by controlling σ_i, thereby guiding t-SNE to consider the local structure of the data. Perplexity is a value defined by Shannon entropy, as follows:

$$Perp(P_i) = 2^{-\sum_j p_{j|i} \log_2 p_{j|i}},$$

where P_i is probability density function in high dimension. The σ_i‘s are estimated corresponding to a given perplexity (refer to Maaten and Hinton (2008) for more detail).

Finally, t-SNE represents the disparity between these distributions using KL-divergence given by

$$Cost = KL(P|Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}},$$

where P and Q are joint probability distributions in the high- and low-dimensional space, respectively. Through gradient descent, t-SNE seeks to minimize the KL-divergence to find low-dimensional data points that best represents the original data. The gradient is

$$\frac{\delta C}{\delta \mathbf{Y}_i} = 4 \sum_j (p_{ij} - q_{ij})(\mathbf{Y}_i - \mathbf{Y}_j)\left(1 + \|\mathbf{Y}_i - \mathbf{Y}_j\|^2\right)^{-1}. \tag{3}$$

Using the momentum method, the parameters are updated by

$$\Gamma^{(m)} = \Gamma^{(m-1)} + \eta \frac{\delta C}{\delta \Gamma} + \alpha(m)\left(\Gamma^{(m-1)} - \Gamma^{(m-2)}\right), \tag{4}$$

where $\Gamma^{(m)}$ is the solution, η represents the learning rate, and $\alpha(m)$ indicates the momentum at iteration m.

3. DTW-based t-SNE for trajectory data

Considering the characteristics of the trajectory data in this study, we propose a joint probability distribution in the high-dimensional space as follows:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n},$$

where the conditional probability is given by

$$p_{j|i} = \frac{exp(-DTW(\mathbf{X}_i, \mathbf{X}_j)/2\sigma_i^2)}{\sum_{i \neq j} exp(-DTW(\mathbf{X}_i, \mathbf{X}_j)/2\sigma_i^2)}. \tag{5}$$

By substituting DTW distance for Euclidean distance in Eq. (1), we achieve a more accurate measurement of similarities, even when dealing with trajectories of different lengths. This newly proposed joint probability distribution in the high-dimensional space ensures a more precise preservation of data structure. In the low-dimensional space, the joint probability distribution is consistently defined with Eq. (2).

Moreover, according to Hinton and Roweis (2002) and Maaten and Hinton (2008), the performance of SNE/t-SNE is fairly robust to changes in perplexity, with recommended values falling between 5 and 50. However, it is noteworthy that even for the same dataset, t-SNE executions yield different low-dimensional representations by varying the perplexities. To address this variability, we considered a set of 10 arbitrary perplexity values, ranging from 5 up to a maximum of $(n-1)/3$. From this set, we suggest selecting the perplexity value that minimizes the total costs (KL-divergence). Algorithm 1 outlines the proposed approach, exploring optimal perplexity and integrating DTW to the original t-SNE for trajectory data adaptation. In Algorithm 1, k indicates the k^{th} perplexity in perplexity vector $Perp$, $\Gamma(t)$ is the solution, η represents the learning rate, and $\alpha(t)$ indicates the momentum at iteration t.

Algorithm 1. DTW-based t-SNE

Data: trajectory data set $\mathbf{X} = \{\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_n\}$
cost function parameters: cost vector $\mathbf{Cost} = \{cost_1, cost_2, ..., cost_k\}$, perplexity vector Perp
optimization parameters: number of iteration M, learning rate η, momentum $\alpha(m)$.
Result: low-dimensional data representation $\Gamma^{(m)} = \{\mathbf{Y}_1, \mathbf{Y}_2, ..., \mathbf{Y}_n\}$.
begin
    Calculate distance matrix D using DTW
    Initialize perplexity vector $\mathbf{Perp} = \{perp_1, perp_2, ..., perp_k\}$
    Set cost vector $\mathbf{Cost} = \{cost_1, cost_2, ..., cost_k\} = \mathbf{0}$
    for $k = 1$ to K **do**
        Execute t-SNE using D with a perplexity $perp_k$
        Calculate the total cost (KL-divergence) $cost_k$
    end
    Choose perplexity $Perp \leftarrow perp_k$ at the minimum $cost_k$
    Compute conditional probability $p_{i|j}$ with perplexity $Perp$
    Set $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$
    Sample initial solution $\Gamma^{(0)} = \{\mathbf{Y}_1, \mathbf{Y}_2, ..., \mathbf{Y}_n\}$ from $N(0, 10^{-4}\mathbf{I})$
    for $m = 1$ to M **do**
        Compute low-dimensional joint probability q_{ij} (2)
        Compute gradient $\frac{\delta C}{\delta \Gamma}$ (3)
        Set $\Gamma^{(m)} = \Gamma^{(m-1)} + \eta \frac{\delta C}{\delta \Gamma} + \alpha(m)(\Gamma^{(m-1)} - \Gamma^{(m-2)})$ (4)
    end
end

4. Numerical study

We evaluated the performance of the newly proposed t-SNE visualization algorithm on trajectory data under various experimental designs by comparing it with two alternatives, traditional t-SNE with Euclidean distance and FD. We considered three groups following three different probability density functions $\phi_X(t)$ in Table 1, each with an equal number of observations. The number of observations considered was $n = \{90, 180, 270\}$, with each observation representing a trajectory, and the length of each trajectory being $l = \{50, 100\}$. Further, for different lengths, we considered the missing rate ϕ and missing time points were randomly selected in either 30% or 50% of the total trajectories in the entire dataset. Figure 2 illustrates examples for the missing rates $\phi = 0\%$ and $\phi = 30\%$.

The performances of visualization results for the three methods are evaluated in terms of four statistics: Wilk's lambda, Pillai's trace, Hotelling–Lawley's trace, and Accuracy. Lower values of Wilk's lambda and higher values of Pillai's trace, Hotelling's trace, and Accuracy indicate better visualization outcomes. Accuracy was assessed by logistic regression model for two groups and K-means clustering for more than three groups using the resulting low-dimensional scores. A total of 50 iterations were repeated for each case, and the average (mean) and standard deviation (sd) for each statistic over the 50 iterations are reported in Table 2.

First, examining the results in terms of three multivariate analyses of variance measures, when the missing rate $\phi = 0$, Euclidean generally outperforms in cases with a sufficient number of samples. However, in scenarios with a small sample size, Fréchet demonstrates superior performance. On the other hand, as the missing rate increases and Euclidean distance becomes undefined, focusing on Fréchet and DTW reveal that the performance of Fréchet deteriorates more significantly than DTW. Consequently, DTW outperforms Fréchet.

Moving on to the accuracy aspect, regardless of the missing rate, DTW consistently exhibits higher accuracy, outperforming both Euclidean and Fréchet. Previous studies (Sakoe and Chiba 1978) have reported that DTW exhibits outstanding performance even in the presence of missing values, and this fact has been reaffirmed through the experiments conducted in this study.

5. Two real-data applications

In this section, we apply the proposed method to two real data, Gait data and NBA data. Since both data have multi-dimensional trajectory structure and so dependency among variables, we considered DTW $_D$ as the distance measure for both datasets. In addition, we applied z-normalization to each variable to ensure consistent scaling and comparability across measurements (Mueen and Keogh 2016).

5.1. Gait data

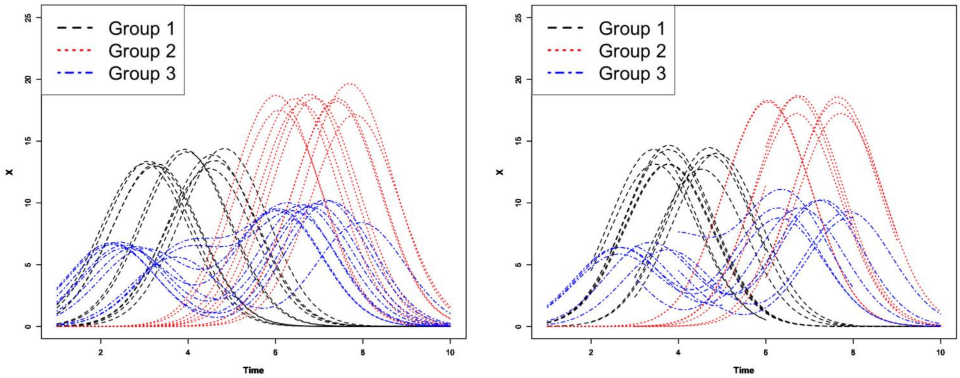
Gait data consists of measurements for the angle of each joint during the walking cycle (1–100%) of a total of 833 pedestrians. It consists of 500 individuals in the normal group and 333 individuals in the abnormal group, further categorized into severity levels ranging from 1 to 4. A pedestrian has a total of 12 joints, with 6 pairs on each side, left and right. As the differences between joints within each pair are not significant, the average of each pair of joints was calculated,

Table 1. The probability density functions $\phi_X(t)$ of three groups.

Group 1	$\phi_X(t) \times Y, X \sim N(4 + U, 1), Y \sim N(35, 2^2)$
Group 2	$\phi_X(t) \times Y, X \sim N(7 + U, 1), Y \sim N(45, 2^2)$
Group 3	$\phi_X(t) \times Y, X \sim 0.4 \times N(3 + U, 1) + 0.6 \times N(7 + U, 1), U \sim U(-1, 1), Y \sim N(40, 2^2)$

Table 2. The performance comparisons of three methods, t-SNE using Euclidean, Fréchet and DTW distance, in multivariate analysis of variance (MANOVA) statistics: Wilk's lambda, Pillai's trace, and Hotelling–Lawley's trace and accuracy based on K-means clustering. Bold numbers represent superior performance for each statistic.

			Wilk's lambda						Pillai's trace						Hotelling–Lawley's trace						Accuracy					
			Euclidean		Fréchet		DTW		Euclidean		Fréchet		DTW		Euclidean		Fréchet		DTW		Euclidean		Fréchet		DTW	
ϕ	n	l	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd	Mean	sd
0	90	50	0.401	0.003	**0.303**	0.003	0.308	0.003	0.599	0.003	**0.697**	0.003	0.692	0.003	11.661	0.465	**33.627**	0.585	12.296	0.214	**0.895**	0.002	0.881	0.002	0.836	0.002
		100	0.295	0.003	**0.240**	0.003	0.333	0.003	0.705	0.003	**0.760**	0.003	0.667	0.003	10.416	0.199	**28.704**	0.547	15.399	0.365	**0.894**	0.002	0.791	0.003	0.855	0.002
	180	50	**0.311**	0.001	0.397	0.001	0.621	0.001	**0.689**	0.001	0.603	0.001	0.379	0.001	**8.089**	0.053	5.908	0.062	0.818	0.006	0.884	0.001	0.875	0.001	**0.922**	0.001
		100	**0.383**	0.001	0.454	0.001	0.595	0.001	**0.617**	0.001	0.546	0.001	0.405	0.001	**5.756**	0.049	5.272	0.063	1.348	0.013	**0.931**	0.001	0.912	0.001	0.903	0.001
	270	50	**0.262**	0.001	0.462	0.001	0.646	0.001	**0.738**	0.001	0.538	0.001	0.354	0.001	**9.766**	0.057	3.599	0.033	0.678	0.002	0.825	0.001	0.882	0.001	**0.912**	0.001
		100	**0.349**	0.001	0.458	0.001	0.628	0.001	**0.651**	0.001	0.542	0.001	0.372	0.001	**5.644**	0.028	5.271	0.049	1.248	0.011	0.892	0.001	0.900	0.001	**0.941**	0.001
30	90	50	–	–	**0.354**	0.002	0.489	0.003	–	–	**0.646**	0.002	0.511	0.003	–	–	**3.498**	0.060	2.494	0.044	–	–	0.883	0.002	**0.921**	0.002
		100	–	–	**0.277**	0.003	0.383	0.003	–	–	**0.723**	0.003	0.617	0.003	–	–	**19.617**	0.331	7.870	0.158	–	–	0.830	0.002	**0.884**	0.002
	180	50	–	–	0.590	0.001	**0.553**	0.001	–	–	0.410	0.001	**0.447**	0.001	–	–	0.742	0.002	**0.933**	0.003	–	–	0.908	0.001	**0.988**	0.001
		100	–	–	**0.528**	0.001	0.625	0.001	–	–	**0.473**	0.001	0.375	0.001	–	–	**1.316**	0.008	0.834	0.006	–	–	**0.966**	0.001	0.932	0.001
	270	50	–	–	0.593	0.001	**0.500**	0.001	–	–	0.407	0.001	**0.500**	0.001	–	–	0.726	0.001	**1.171**	0.002	–	–	0.907	0.001	**0.996**	0.001
		100	–	–	**0.529**	0.001	0.604	0.001	–	–	**0.471**	0.001	0.396	0.001	–	–	**1.908**	0.013	0.812	0.002	–	–	0.920	0.001	**0.931**	0.001
50	90	50	–	–	0.476	0.002	**0.457**	0.002	–	–	0.524	0.002	**0.543**	0.002	–	–	1.488	0.018	**2.266**	0.033	–	–	0.898	0.002	**0.932**	0.002
		100	–	–	**0.357**	0.002	0.439	0.003	–	–	**0.643**	0.002	0.561	0.003	–	–	**5.814**	0.129	3.729	0.064	–	–	**0.911**	0.002	0.909	0.002
	180	50	–	–	0.629	0.001	**0.532**	0.001	–	–	0.371	0.001	**0.468**	0.001	–	–	0.598	0.001	**0.966**	0.002	–	–	0.947	0.001	**0.992**	0.001
		100	–	–	0.578	0.001	**0.564**	0.001	–	–	0.422	0.001	**0.436**	0.001	–	–	0.790	0.002	**0.988**	0.004	–	–	0.920	0.001	**0.969**	0.001
	270	50	–	–	0.645	0.001	**0.515**	0.001	–	–	0.355	0.001	**0.485**	0.001	–	–	0.555	0.001	**1.050**	0.002	–	–	0.943	0.001	**0.992**	0.001
		100	–	–	0.594	0.001	**0.566**	0.001	–	–	0.406	0.001	**0.434**	0.001	–	–	0.731	0.001	**0.924**	0.002	–	–	0.881	0.001	**0.989**	0.001



(a) n=30, l=50, ϕ=0% (b) n=30, l=50, ϕ=30%

Figure 2. The sample data consists of 10 trajectories per group (total 30 trajectories), each comprising 50 time points. (a) 30 trajectories without missing values and (b) 30 trajectories with 30% missing values, having different lengths.

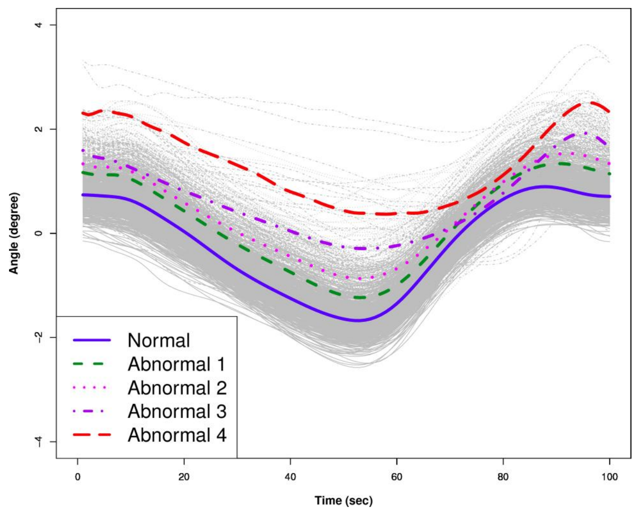
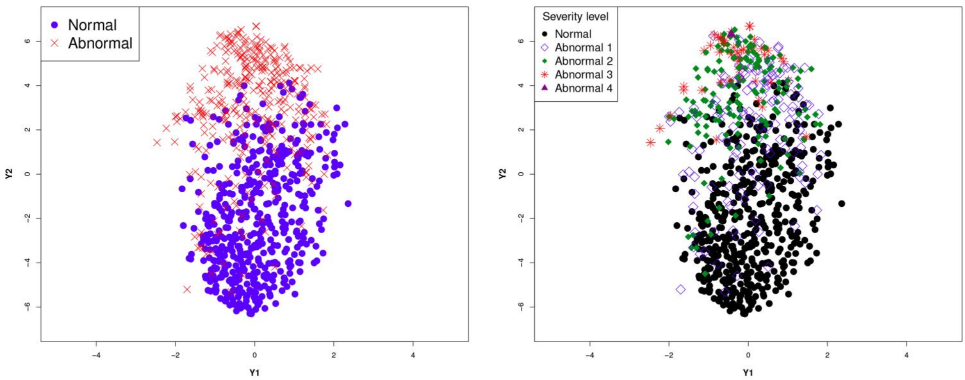


Figure 3. The angles trajectories of joint J1 in 833 pedestrians over time, along with the mean curves for each group.

resulting in a total analysis of 6 joints (J1–J6). For instance, Figure 3 displays the trajectories for a single joint J1, showing changes in the mean curve as severity increases. (See Appendix A1 for the mean curves of other joints. This data used in this analysis cannot be disclosed due to the request of the data provider.)

We performed proposed DTW-based t-SNE visualization using the proposed Algorithm, considering each joint individually from J1 to J6, and Figure 4 presents the DTW-based t-SNE result of J1 according to (a) group and (b) severity level and Table 3 shows its accuracy for all six joints, calculated as the average over 100 iterations with different initial values for a given joint. Among these, the low-dimensional embedding of trajectories of joint J1 and J2 exhibited the highest accuracy, and we ultimately selected these two joints for the multivariate trajectory data afterward. Note that higher accuracy implies that the labels of actual normal and abnormal groups were well reflected in the visualizations. The visualization results for other joints (J3–J6) can be found in Appendix A2. Through the visualization of individual variables (joints), we observed that the differences in the shape of the mean curves by group were reflected in Figure 4a. By simultaneously considering both J1 and J2 joints using mDTW, Figure 5a exhibits slightly

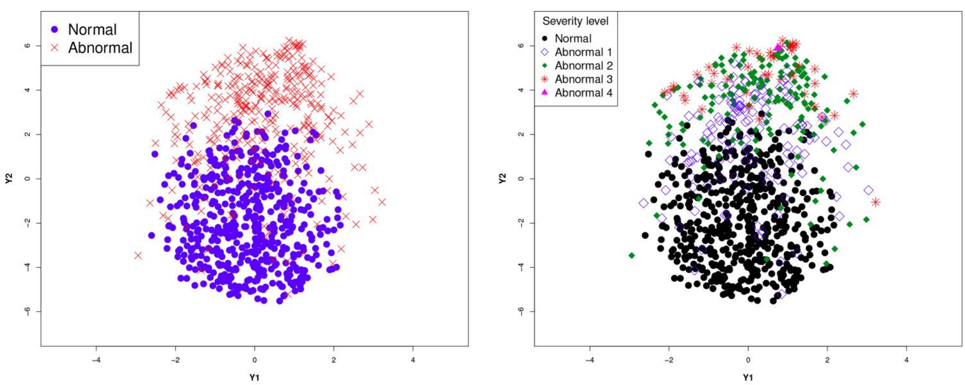


(a) (b)

Figure 4. The visualization of DTW-based t-SNE results considering joint J1. (a) Plot according to two groups: normal and abnormal groups (accuracy: 0.852). (b) Plot according to severity level from 1 to 4.

Table 3. Accuracy for t-SNE visualization of each six joints.

Joint	J1	J2	J3	J4	J5	J6
Accuracy	0.852	0.868	0.792	0.716	0.719	0.699



(a) (b)

Figure 5. The visualization of DTW-based t-SNE results considering both J1 and J2 joints, simultaneously. (a) Plot according to two groups: the normal and the abnormal group (accuracy: 0.878). (b) Plot according to severity levels from 1 to 4.

enhanced group separation compared to Figure 4a with increasing accuracy. Furthermore, Figure 5b effectively capture the gradual changes in the trajectory characteristics of gait data with increasing severity, showing a slight improvement compared to Figure 4b.

5.2. NBA data

The dataset records the trajectory of successful 3-point shots made by 128 NBA players from 2013 to 2016. The data was collected to explore different perspectives on the trajectories of 3-point shots and aims to enhance understanding of the necessary skills and strategies employed by basketball players for successful 3-point shots. Figure 6 illustrates the plotted trajectories of the ball for 9 key players from overhead views.

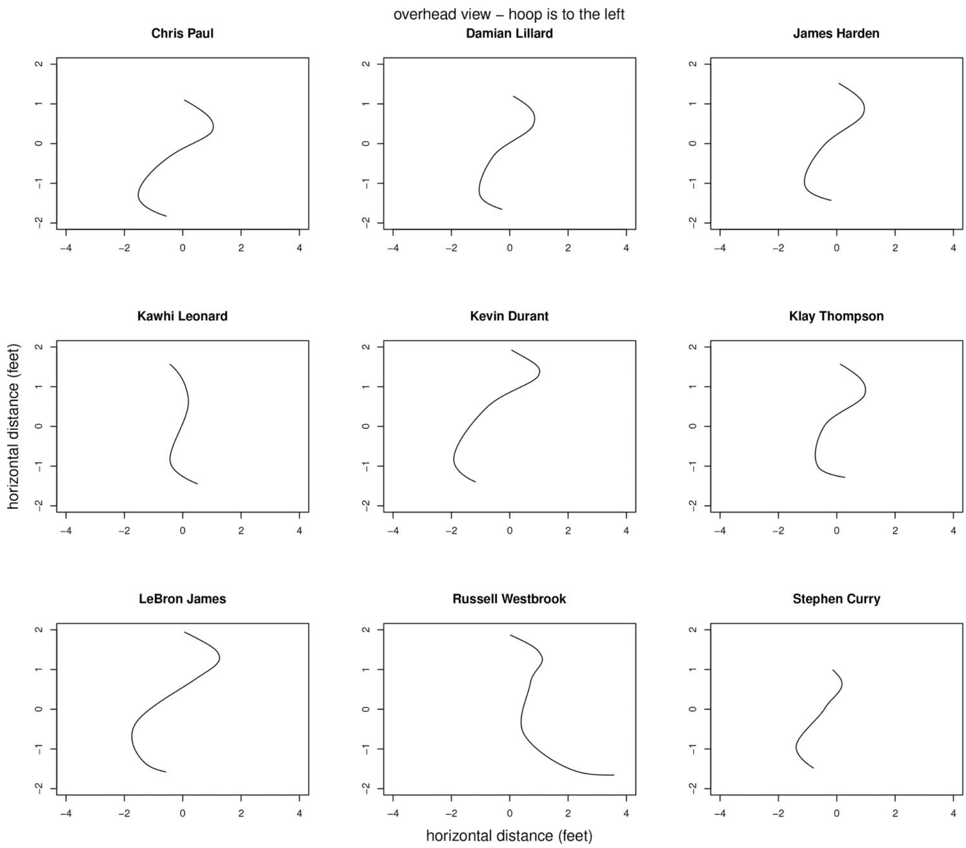


Figure 6. The overhead view for 9 key players during release time, taking into account the x-coordinate and z-coordinate.

The ball's trajectory was measured in three directions based on release time. The x-coordinate denotes left/right positioning from the player's perspective, the y-coordinate indicates the distance from the basketball hoop, and the z-coordinate represents the height of the ball. In the analysis, two variables, x-axis and z-axis, were considered as depicted in Figure 6. Additional comprehensive information about the data is available in the provided source, NBA player shooting motions.[1]

We applied the new DTW-based t-SNE method to the NBA data, considering two variables, x- and z-axes. Before delving into dimension reduction, we observed that most trajectories were densely clustered around the center, while specific players exhibited a tendency to throw the ball either to the left or right. Initially, we expected that players might display different shot directions based on their rankings. However, upon examining the DTW-based t-SNE result, it was challenging to discern distinct patterns and NBA rankings were not solely determined by 3-point shots.

Further, to explore the low-dimensional scores resulting from DTW-based t-SNE, we employed K-means clustering, an unsupervised learning algorithm that groups data into K clusters. To determine the appropriate K value, we considered K values from 1 to 10, and visually examined the variation in cluster numbers to find the point where the decrease in variation becomes less steep, forming an "elbow" shape. Based on this analysis, a K value of 4 was chosen to group the data into four clusters and these labels are plotted in Figure 7. Analyzing the average trajectory curves of 128 players in Figure 8 reveals that NBA players employ four different strategies when attempting 3-pointers.

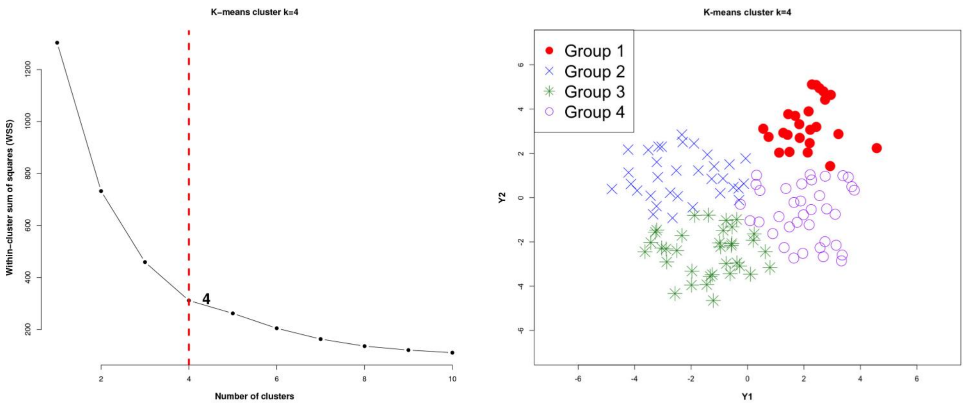


Figure 7. *K*-means clustering with $K = 4$ for the low-dimensional embedding for NBA data by DTW-based t-SNE.

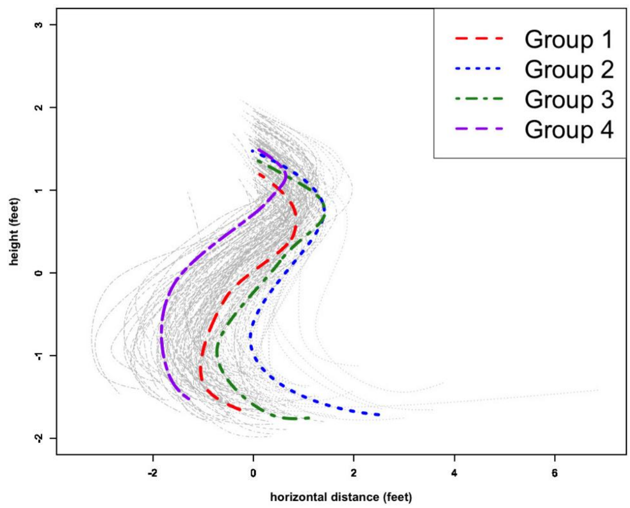


Figure 8. The trajectory of the ball for 128 players with the mean curves of the four strategies determined by the *k*-means clustering using the low-dimensional embedding by DTW-based t-SNE.

6. Conclusion

This study introduces a novel approach utilizing DTW in t-SNE for the visualization and analysis of trajectory data, as demonstrated through a simulation study and two real data applications, gait data and NBA data. The proposed method effectively addresses the challenges posed by trajectory data, accommodating variations in length and observed points. DTW plays a pivotal role, providing accurate similarity measurements even when trajectories exhibit different lengths and observed points.

The simulation study demonstrates the robustness and versatility of the proposed approach under different experimental designs and conditions. The comparison with traditional metrics like Euclidean and FDs highlights the superiority of DTW-based t-SNE, particularly in handling the increased complexity of trajectory data. In the analysis of gait data, by applying the proposed method to joints, the study successfully captures severity-specific gait patterns. In the context of NBA 3-point shot trajectories, the application of DTW-based t-SNE with the K-means clustering results in the four patterns. These applications showcase the effectiveness of DTW-base t-SNE in revealing subtle differences and similarities in multivariate trajectory dataset.

In conclusion, the presented DTW-based t-SNE method emerges as a powerful tool for researchers and analysts working with trajectory data. The combination of DTW and t-SNE offers a nuanced and accurate means of visualizing and interpreting complex spatiotemporal patterns.

As trajectory data become increasingly prevalent in various domains, this approach opens new avenues for deeper exploration and understanding of dynamic phenomena.

Disclosure statement

The authors declare no potential conflict of interest.·

Note

1. https://www.inpredictable.com/2021/01/nba-player-shooting-motions-data-dump.html

Funding

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) grant funded by the Ministry of Education (NRF-2021R1A6A1A10044950) and Global – Learning & Academic research institution for Master's · PhD students, and Postdocs (G-LAMP) Program of the National Research Foundation of Korea (NRF) grant funded by the Ministry of Education (No. RS-2023-00285390).

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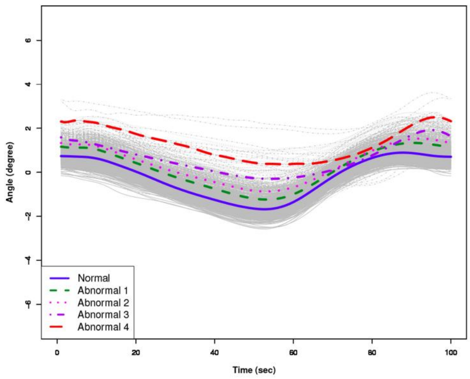
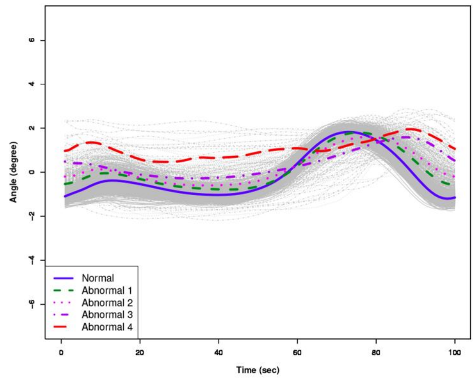
Appendix

Appendix A Gait data

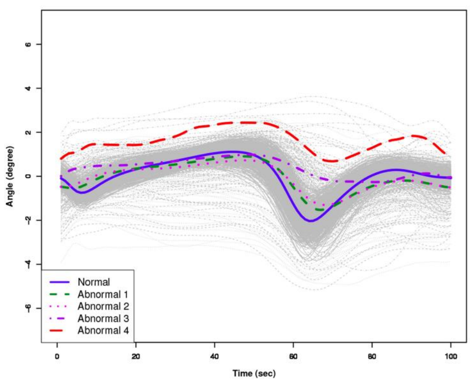
A1. trajectories for six joints



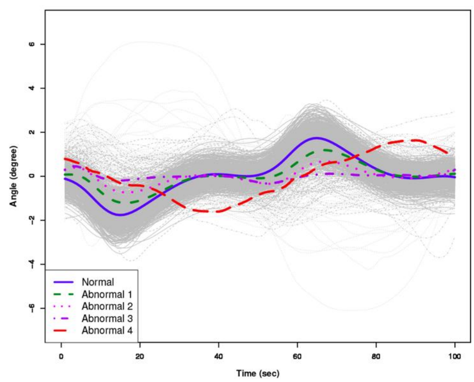
(a) J1



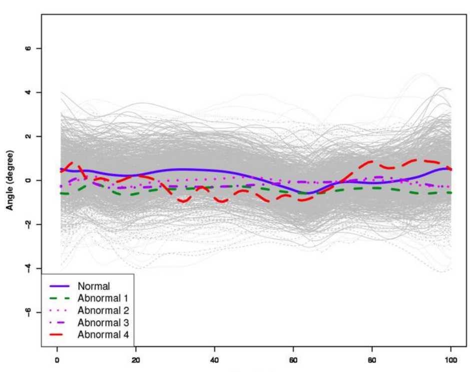
(b) J2



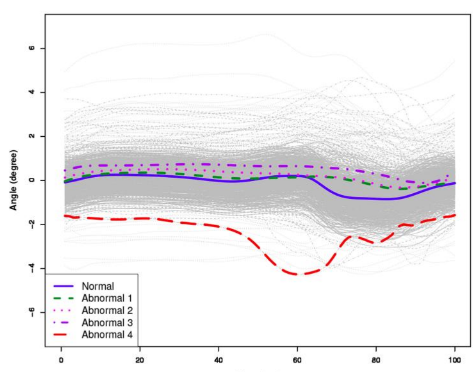
(c) J3



(d) J4



(e) J5



(f) J6

Figure A1. The angle trajectories for six joints in 833 pedestrians over time, along with the mean curves for each group.

A2. DTW-based t-SNE visualization

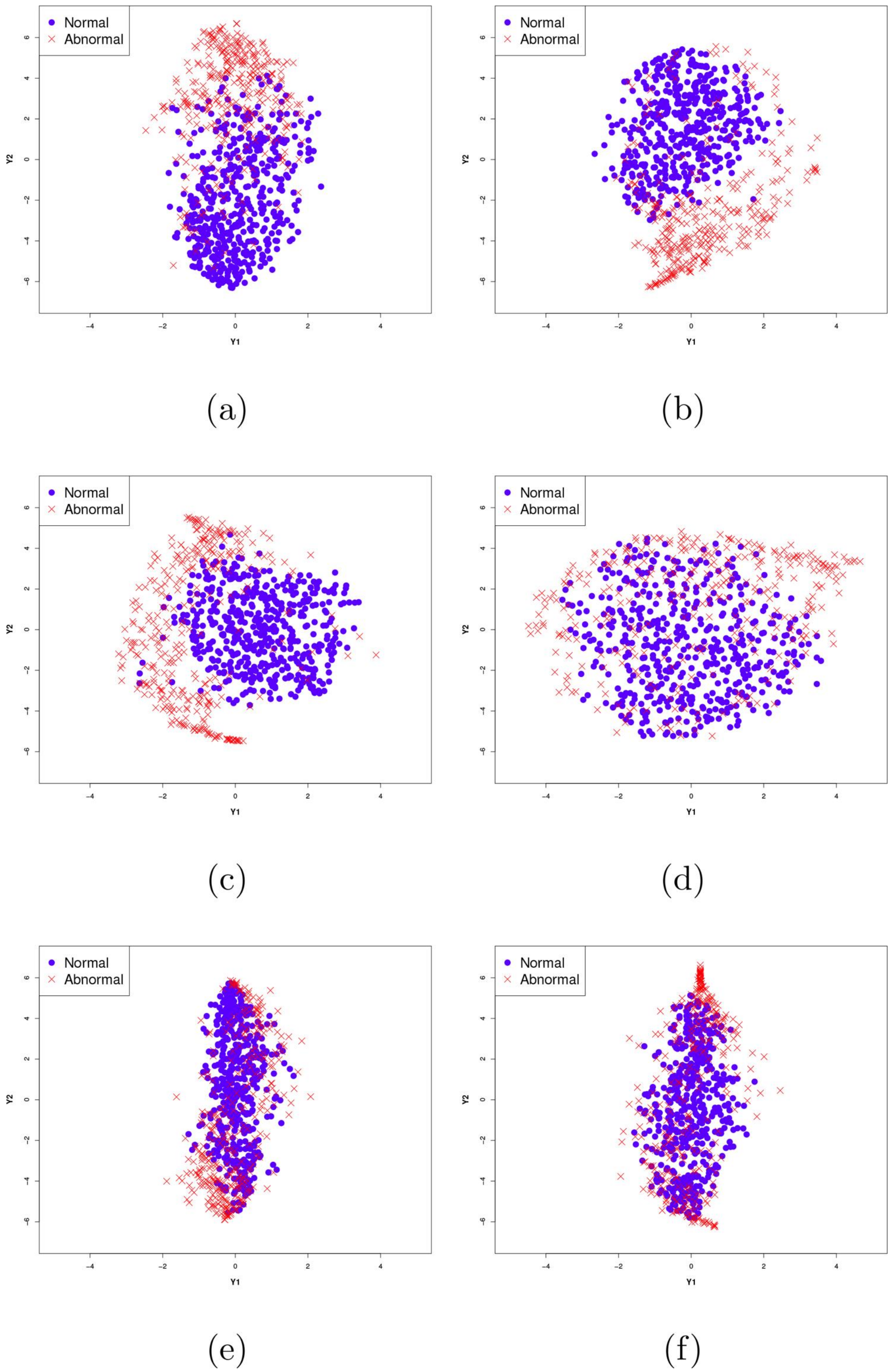


Figure A2. The visualization of DTW-based t-SNE results considering each joint individually. (a) joint J1 (accuracy: 0.852), (b) joint J2 (accuracy: 0.868), (c) joint J3 (accuracy: 0.792), (d) joint J4 (accuracy: 0.716), (e) joint J5 (accuracy: 0.719), (f) joint J6 (accuracy: 0.699), with accuracy calculated by logistic model.